PROJECT WORK

Name: CHINMAY MOHAPATRA

REG. NO: 25BAI10986

Branch: B.Tech CSE (AI/ML)

Year: 1st YEAR

Course Code: CSA2001

Course Title: Fundamentals of Ai & Ml

Faculty: Dr. VAISHNAVI S.

Date: 30 /03 /26

1. Introduction

AI Wealth Visualiser is a terminal-based intelligent financial planning system designed to help users understand, manage, and forecast their financial future. It integrates Artificial Intelligence, financial mathematics, and simulation techniques to create realistic investment scenarios.

The system transforms raw financial inputs into actionable insights using data-driven models. Instead of relying on fixed assumptions, it uses probabilistic analysis to reflect real-world uncertainty, enabling better decision-making in an increasingly volatile economic landscape.

2. Objectives

The project focuses on delivering a structured and intelligent financial analysis system.

- **Financial Analysis:** Evaluate income, expenses, and savings patterns.
- **Wealth Forecasting:** Predict long-term wealth growth through compounding.
- **Risk Assessment:** Simulate uncertainty using probabilistic models to determine the "Value at Risk."
- **Goal Planning:** Align financial decisions with target wealth using back-testing logic.

3. System Architecture

The system follows a **Pipeline Architecture**, ensuring a unidirectional flow of data that maintains integrity and modularity.

1. **Ingestion Layer:** Validates and sanitizes user-provided financial metrics.
2. **Logic Layer:** Executes the core financial formulas and applies the selected risk profile.
3. **Stochastic Layer:** The AI core where Monte Carlo simulations introduce market variance.
4. **Presentation Layer:** Uses ANSI-coded graphics to render terminal-based visual representations of data.

---

4. Functional Modules

The system is divided into multiple modules:

Input & Validation Module

- **Role:** Collects user financial data (Income, Expenses, Age, Target).
- **Feature:** Implements error-handling for non-numeric inputs and unrealistic financial ratios (e.g., expenses exceeding income).

Simulation Engine (The AI Core)

- **Role:** Runs N iterations of financial life paths.

- **Logic:** Uses a random walk model where each year's return is sampled from a normal distribution defined by the user's risk profile.

---

5. AI Techniques & Mathematical Modelling

Monte Carlo Simulation

Unlike a standard calculator, the system acknowledges that market returns are not constant. We model the annual return R_t as a random variable:

$$R_t \sim \mathcal{N}(\mu, \sigma^2)$$

Where μ is the expected mean return and σ is the volatility (standard deviation).

State Space Representation

The system treats the financial journey as a sequence of states S:

$$S_t = \{Age, Capital, Annual_Savings\}$$

The transition function $f(S_t) \rightarrow S_{t+1}$ accounts for inflation, tax implications, and reinvested dividends.

Search-Based Optimization

By utilizing a **Breadth-First Search (BFS)** approach in the simulation space, the agent explores various "what-if" scenarios (e.g., "What if I retire 5 years early?") to find the optimal path to the goal state.

---

6. Intelligent Agent Model

The system operates as a **Utility-Based Agent**, aiming to maximize the utility function U(W), where W is the final wealth.

Component	Description
Agent Type	Goal-Based + Utility-Based
Environment	Stochastic financial markets
Perception	User savings rate, risk tolerance, time horizon
State	Current Net Worth + Portfolio Allocation
Action	Rebalancing, adjusting SIP, or changing risk profiles
Utility	Maximizes probability of hitting $Goal$ while minimizing $Downside Risk$

7. Core Algorithms

Compound Interest (Lump Sum)

To calculate the future value of an initial investment:

$$A = P(1 + r)^n$$

Systematic Investment Plan (SIP)

For recurring monthly contributions:

$$FV = P \times \frac{(1 + r)^n - 1}{r} \times (1 + r)$$

Normal Distribution Sampling

For every year in a simulation, the return is calculated using the Box-Muller transform or similar methods to simulate market "noise":

$$Return = \mu + (Z \times \sigma)$$

(Where Z is a random standard normal variable)

8. Investment Profiles & Risk Parameters

The system uses predefined statistical parameters for different investor types:

Profile	Expected Return (μ)	Volatility (σ)	Asset Mix
Conservative	5%	2%	Bonds / Cash
Moderate	8%	7%	Index Funds / Blue Chip
Aggressive	12%	15%	Growth Stocks / Tech
High-Risk	18%	25%	Crypto / Emerging Markets

9. Complexity & Performance

- **Time Complexity:** O(S X Y), where S is the number of simulations (e.g., 1000) and Y is the number of years.
- **Space Complexity:** O(S X Y) to store the paths for rendering the final percentile charts.
- **Optimization:** The system utilizes **Threading** to run simulations in parallel, significantly reducing computation time on multi-core processors.

---

10. Output Interpretation & Insights

The system provides a **Confidence Interval** for the user's goal:

- **90th Percentile (Best Case):** "The Bull Market scenario."
- **50th Percentile (Median):** The most likely outcome.
- **10th Percentile (Worst Case):** The "Safety Net" figure.

Success Rate Calculation:

$$Success\% = \frac{\text{Simulations where Final Wealth} \geq \text{Target Wealth}}{\text{Total Simulations}} \times 100$$

---

11. Detailed Logic & Mathematical Proofs

To ensure the system is more than just a "calculator," we implement **Geometric Brownian Motion (GBM)** concepts to model wealth.

Stochastic Differential Equation

The change in wealth (dW_t) over a small time increment (dt) is modelled as:

$$dW_t = \mu W_t dt + \sigma W_t dW_t + Sdt$$

- $\mu W_t dt$: The deterministic growth (drift).
- $\sigma W_t dW_t$: The random market shocks (diffusion).
- Sdt: The continuous savings rate injected into the system.

By discretizing this for a terminal-based system, we calculate each year as:

$$W_{t+1} = (W_t + S) \times e^{(\mu - 0.5\sigma^2) + \sigma Z}$$

(Where Z is a random variable sampled from a Standard Normal Distribution).

---

12. System Design (Class Hierarchy)

The system is built using **Object-Oriented Programming (OOP)** to ensure the AI Agent can be easily swapped or upgraded.

Core Classes

1. **Finance Engine:** The "Model." Contains all the pure math formulas (FV, PV, SIP).
2. **Simulator:** The "Controller." Manages the loops for Monte Carlo iterations and handles threading.
3. **Investor Agent:** The "Intelligence." Holds the state (Risk Profile) and makes "decisions" on how to rebalance the portfolio if the simulation trends toward failure.
4. **Visualizer:** The "View." Converts raw float arrays into ANSI bar charts and tables.

---

13. Simulation Methodology: The "Fan Chart"

Instead of a single line graph, the AI Wealth Visualiser generates a **Fan Chart** (Probability Density).

- **The Median Path (P50):** Represents the "Expected" outcome.
- T**he Confidence Interval (P5th to P95th):** The shaded area representing 90% of all possible futures.
- **The "Black Swan" Event:** The bottom 1% of simulations, helping users prepare for extreme market crashes.

---

14. Comparative Analysis of Risk Profiles

The system uses the **Sharpe Ratio** logic to differentiate between profiles.

Metric	Conservative	Moderate	Aggressive
Asset Allocation	80% Debt / 20% Equity	50% Debt / 50% Equity	10% Debt / 90% Equity
Max Drawdown	-5% to -10%	-20% to -25%	-40% to -60%
Recovery Time	1-2 Years	3-5 Years	7+ Years
Primary Goal	Capital Preservation	Balanced Growth	Wealth Creation

---

15. AI Agent Decision Logic (Pseudo-Code)

The "Intelligent" part of the agent uses a heuristic to suggest mid-course corrections during a simulation:

Python

```
IF Success_Rate < 70%:
  ADVICE = "Increase monthly savings by " + Calculate_Gap()
ELIF Portfolio_Volatility > User_Threshold:
  ADVICE = "Rebalance toward Conservative assets"
ELSE:
  ADVICE = "On track. Maintain current strategy."
```

---

16. Error Handling and Data Integrity

To maintain "Production-Grade" reliability, the system implements:

- **Inflation Normalization:** All future values are discounted back to "Today's Value" using a standard 4-6% inflation rate, preventing "Nominal Illusion."
- **Boundary Constraints:** Prevents the simulation from running if the user's "Withdrawal Rate" exceeds a sustainable threshold (The 4% Rule).
- **Seed Control:** Allows users to input a "Simulation Seed" to replicate the exact same "random" market conditions for comparative testing.

---

17. Evaluation Metrics

We evaluate the system's performance based on:

1. **Convergence Stability:** Does the Success Rate stabilize after 10,000 iterations?
2. **Computation Latency:** Time taken to process 50 years of data across 1,000 scenarios (Target: < 2.0 seconds).
3. **Predictive Realism:** How closely the simulated standard deviation matches historical S&P 500 or NIFTY 50 data.

---

18. Societal Impact & Ethics

- **Financial Literacy:** Democratizes high-end financial modelling typically reserved for private wealth management clients.

- **Bias Mitigation:** The AI does not "prefer" any specific financial product; it remains mathematically objective.
- **Risk Transparency:** Explicitly shows the user the "Probability of Ruin," discouraging reckless financial behaviour.

---

19. Scalability and Modular Extensions

The system is designed with an **Open-Closed Principle**:

- **Tax Modules:** New modules for specific country tax codes (e.g., 401k, ISA, 80C) can be "plugged in" to the processing layer without altering the simulation engine.
- **API Layer:** While currently terminal-based, the Simulation Engine class is decoupled from the Visualizer, allowing it to serve as a backend for a **Web-based Dashboard** or **Mobile App** in the future.

---

19. Summary of System Advantages

- **Non-Linear Thinking:** Moves the user away from "8% every year" to a "Range of possibilities."
- **Low Overhead:** Runs on any machine with Python—no cloud or expensive GPU required.
- **Objective Analysis:** Eliminates emotional bias by showing the mathematical probability of failure.

---

20. Conclusion of the Extended Report

The **AI Wealth Visualiser** represents a shift from static financial planning to **Dynamic Financial Intelligence**. By combining the rigor of stochastic calculus with a lightweight, accessible terminal interface, it provides a high-fidelity tool for long-term wealth management. It is not just a tool for calculation, but a tool for **risk-awareness**, enabling users to navigate the uncertainty of the future with mathematical confidence.